

A norm-null/ c_0 border for Question 5.15 of Martínez–Meza–Alcántara–Uzcátegui

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Source question

The source is Jorge Martínez, David Meza-Alcántara, and Carlos Uzcátegui, *Pathology of submeasures and F_σ ideals*, arXiv:2211.01544v2. After proving that, for sequences in c_0 ,

$$\mathcal{R} \leq_K \mathcal{B}(\mathbf{x}) \iff (x_n)_n \text{ is weakly null,}$$

the authors ask:

Which Banach spaces satisfy that $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$ for every weakly null sequence $\mathbf{x} = (x_n)_n$?

Theorem 5.14. *Let $\mathbf{x} = (x_n)_n$ be a sequence in c_0 . Then $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$ iff $(x_n)_n$ is weakly null.*

Proof. If $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$, then $\mathcal{B}(\mathbf{x})$ is tall and thus $\mathbf{x} = (x_n)_n$ is weakly null by Proposition 5.13. Conversely, let $(x_n)_n$ be a weakly null sequence in c_0 . By Lemma 5.7, we can assume that $x_n(k) \geq 0$ for all n and k . The corresponding measures are given by $\mu_k(\{n\}) = x_n(k)$. Since each $x_n \in c_0$, $(\mu_k)_k$ is of type c_0 (see definition 4.9). Notice that $\varphi_{\mathbf{x}} = \sup_k \mu_k$ and thus $\mathcal{B}(\mathbf{x}) = \text{FIN}(\varphi_{\mathbf{x}})$. Condition (iii) in Theorem 4.10 (namely, $\lim_n \mu_k(\{n\}) = 0$ for all k) is the translation of being weakly null in c_0 . Thus $\mathcal{R} \leq_K \text{FIN}(\varphi_{\mathbf{x}})$. $\square$

The previous result naturally suggests the following.

Question 5.15. *Which Banach spaces satisfies that $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$ for every weakly null sequence $\mathbf{x} = (x_n)_n$?*

In relation to the previous question. Let $\mathbf{x} = (x_n)_n$ be the usual unit basis of l_2 (which is weakly null). Since l_2 does not contain copies of c_0 , $\mathcal{B}(\mathbf{x}) = \text{FIN}$. Moreover, the same happens in l_∞ , as this space contains isomorphic copies of every separable Banach space.

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Idea of the proof

There are two positive mechanisms. First, if $\|x_n\| \rightarrow 0$, then the submeasure associated to $\mathcal{B}(\mathbf{x})$ has property A in the sense of Section 4 of the source paper. Proposition 4.8 of the source then gives $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$. Thus all Schur spaces satisfy the question.

Second, the source theorem already solves the problem for c_0 . This is stable under passing to subspaces of c_0 , because the ideal $\mathcal{B}(\mathbf{x})$ only records uniform boundedness of finite partial sums, and this is preserved by an isomorphic embedding.

The negative part uses the opposite mechanism. If $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$, then $\mathcal{B}(\mathbf{x})$ must be tall. For a seminormalized weakly null sequence with no c_0 -subsequence, no infinite subseries can be perfectly bounded: otherwise it would be weakly unconditionally Cauchy, and the Bessaga-Pelczynski selection principle would produce a c_0 -subsequence.

Partial answer

Theorem 1. *Let X be a Banach space.*

1. *For every sequence $\mathbf{x} = (x_n)_n$ in any Banach space, if $\|x_n\| \rightarrow 0$, then*

$$\mathcal{R} \leq_K \mathcal{B}(\mathbf{x}).$$

Consequently every Schur space satisfies Question 5.15.

2. *If X is isomorphic to a subspace of c_0 , then for every weakly null sequence $\mathbf{x} = (x_n)_n$ in X one has*

$$\mathcal{R} \leq_K \mathcal{B}(\mathbf{x}).$$

3. *If X contains a seminormalized weakly null sequence with no subsequence equivalent to the unit vector basis of c_0 , then X does not satisfy the property in Question 5.15.*

Proof. For (1), let $\varphi_{\mathbf{x}}$ be the lower semicontinuous submeasure associated to $\mathbf{x}$ in Lemma 5.2 of the source paper:

$$\varphi_{\mathbf{x}}(A) = \sup \left\{ \left\| \sum_{n \in F} x_n \right\| : \emptyset \neq F \in [A]^{<\omega} \right\}.$$

Lemma 5.2 gives $\mathcal{B}(\mathbf{x}) = \text{FIN}(\varphi_{\mathbf{x}})$. If $\|x_n\| \rightarrow 0$, then for every $\varepsilon > 0$ the set

$$M_{\varepsilon} = \{n : \varphi_{\mathbf{x}}(\{n\}) > \varepsilon\} = \{n : \|x_n\| > \varepsilon\}$$

is finite, hence belongs to $\text{FIN}(\varphi_{\mathbf{x}})$. Thus $\varphi_{\mathbf{x}}$ has property A. Proposition 4.8 of the source paper gives

$$\mathcal{R} \leq_K \text{FIN}(\varphi_{\mathbf{x}}) = \mathcal{B}(\mathbf{x}).$$

If X has the Schur property, every weakly null sequence in X is norm-null, so the preceding argument applies to every weakly null sequence.

For (2), let $J : X \rightarrow c_0$ be an isomorphic embedding and let $(x_n)_n$ be weakly null in X . Then $(Jx_n)_n$ is weakly null in c_0 , since every coordinate functional on c_0 , composed with J , is a bounded functional on X . By Theorem 5.14 of the source paper,

$$\mathcal{R} \leq_K \mathcal{B}((Jx_n)_n).$$

It remains only to identify the ideal. Since J is an isomorphic embedding, there are constants $a, b > 0$ such that

$$a\|u\|_X \leq \|Ju\|_{c_0} \leq b\|u\|_X \quad (u \in X).$$

Therefore, for every $A \subseteq \mathbb{N}$,

$$\sup_{F \in [A]^{<\omega}} \left\| \sum_{n \in F} x_n \right\|_X < \infty \iff \sup_{F \in [A]^{<\omega}} \left\| \sum_{n \in F} Jx_n \right\|_{c_0} < \infty.$$

Thus $\mathcal{B}(\mathbf{x}) = \mathcal{B}((Jx_n)_n)$, and the desired Katetov reduction follows.

For (3), let $(x_n)_n$ be seminormalized, weakly null, and have no c_0 -subsequence. We claim that $\mathcal{B}(\mathbf{x})$ is not tall. Indeed, if some infinite $A \subseteq \mathbb{N}$ belonged to $\mathcal{B}(\mathbf{x})$, then the series $\sum_{n \in A} x_n$ would be perfectly bounded. By Proposition 5.12 of the source paper, equivalently,

$$\sum_{n \in A} |x^*(x_n)| < \infty \quad \text{for every } x^* \in X^*,$$

so the subseries is weakly unconditionally Cauchy. Since $(x_n)_{n \in A}$ is seminormalized, the classical Bessaga-Pelczynski c_0 principle yields a subsequence equivalent to the unit vector basis of c_0 , contradicting the choice of (x_n) . Hence $\mathcal{B}(\mathbf{x})$ contains no infinite subset of $\mathbb{N}$, so it is not tall.

The source paper recalls that $\mathcal{R} \leq_K \mathcal{I}$ implies $\mathcal{I}$ is tall. Consequently $\mathcal{R} \not\leq_K \mathcal{B}(\mathbf{x})$, and X fails Question 5.15. $\square$

Corollary 1. *Every infinite-dimensional reflexive Banach space fails the property in Question 5.15.*

Proof. Every infinite-dimensional reflexive Banach space contains a seminormalized weakly null basic sequence. Since a reflexive space contains no copy of c_0 , that sequence has no c_0 -subsequence. Apply Theorem 1(2). $\square$

Verification and limitations

This is a partial answer, not a full characterization. It gives two broad positive classes, Schur spaces and subspaces of c_0 , and a broad obstruction, all spaces containing a seminormalized weakly null sequence without a c_0 -subsequence. The remaining border concerns non-Schur spaces where every seminormalized weakly null sequence has a c_0 -subsequence, but which are not known or not assumed to embed into c_0 . At the sequence level, the issue is that tallness of $\mathcal{B}(\mathbf{x})$ is weaker than $\mathcal{R} \leq_K \mathcal{B}(\mathbf{x})$; the source paper's Question 4.17 asks whether every nonpathological tall F_σ ideal is above $\mathcal{R}$, so this frontier is tied to the paper's broader Katetov problem.

Bounded novelty checks used the run indexes for ‘2211.01544’, ‘ $\mathcal{B}(\mathbf{x})$ ’, weakly null, random ideal, Schur property, and c_0 , plus web searches for the exact wording of Question 5.15 and related phrases. No later explicit answer was found. The nearby run packet on arXiv:2508.21235 concerns critical ideals of ideal Schauder bases and nonpathological Banach-representable ideals; it is related background but does not record this norm-null/ c_0 -border statement for Question 5.15.

References

- [1] J. Martínez, D. Meza-Alcántara, and C. Uzcátegui, *Pathology of submeasures and F_σ ideals*, arXiv:2211.01544.
- [2] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, Springer, 2016.